Fairness in Machine Learning Models for Healthcare and Education

Shubin Li
Engineering&Computing
Dublin City University
Dublin, Ireland
shubin.li2@mail.dcu.ie

Lamaan Kayum Shaikh
Engineering&Computing
Dublin City University
Dublin, Ireland
lamaankayum.shaikh2@mail.dcu.ie

I. Introduction

With the rapid growth of data availability and advances in computational capacity, machine learning models have been increasingly deployed in high-stakes scenarios such as healthcare and education for tasks including disease risk prediction, medical resource allocation, and individual performance assessment [1]. According to the AI Act [2], machine learning systems used for medical prediction and student performance evaluation are classified as high-risk applications, for which relevant regulations explicitly require assessments and safeguards concerning model transparency and fairness.

In the healthcare field, machine learning models have been widely applied to tasks such as depression risk prediction, patient readmission risk assessment, and disease progression analysis [3,4,5,6]. The outputs of these models may directly influence diagnostic decisions and the allocation of medical resources [7]. While such systems demonstrate significant potential for enhancing healthcare efficiency and supporting clinical decision-making, growing evidence suggests that these models may also amplify or reinforce existing social inequalities [8]. Due to differences in data distributions, historical biases, or inadequate methodological design, ML models may exhibit inconsistent predictive behavior across demographic groups, thereby introducing fairness risks [3,6]. Especially in low- and middle-income countries, where algorithmic governance and regulatory mechanisms remain inadequate, AI systems may pose more pronounced fairness risks [8].

These fairness challenges are not unique to healthcare. In the education domain, predictive models are deployed in a similar way to guide institutional decision-making. Machine learning plays a great role in education. Schools use these models to predict things like whether students will struggle, drop out, or succeed usually by analyzing student behavior and background data. With the rise of large learning analytics datasets, researchers can track engagement in detail, looking at everything from test scores to how often students log on to online platforms [9,10]. Predictive techniques logistic regression, decision trees, ensemble methods have focused mainly on getting accurate results and spotting risks early. However, these efforts pay little attention to fairness or ethics [11].As these prediction systems become part of how institutions make decisions, concerns have emerged. Models trained on old data can end up repeating or even worsening existing inequalities. Bias can be introduced through unbalanced datasets, variables that stand in for sensitive characteristics, or by deploying models in ways that hit some groups harder than others [12,13,14,15].

Given that healthcare and education are widely regarded as high-risk application domains and both play a foundational role in shaping individuals' long-term development and social mobility, systematic evaluation of model fairness has become an unavoidable research challenge. In these contexts, algorithmic predictions often directly influence individuals' access to opportunities and the allocation of social resources. Although the two domains differ in context and data characteristics, both present risks of group-level disparities, warranting a cross-domain perspective on algorithmic fairness.

In recent years, substantial research efforts have focused on algorithmic fairness in healthcare and education. Some studies investigate specific disease prediction tasks by analyzing disparities in model predictions across gender or other demographic attributes [6], while others propose and compare various bias detection and mitigation techniques to examine their effects on both predictive performance and fairness [3,4,16]. In addition, the development of general toolkits and evaluation frameworks has provided a range of fairness metrics and mitigation strategies, facilitating fairness analysis across different models and datasets [17].

Despite these advances, several limitations remain. First, fairness evaluations are often confined to specific prediction tasks, making it difficult to assess the stability of fairness methods across different application scenarios [4]. Second, prior studies typically examine bias detection and mitigation in isolation, with limited consideration of the integrated "detection–mitigation–reassessment" workflow, thereby constraining practical applicability [3,4,6]. Finally, although some methods are designed to be model-agnostic, their reusability across domains still lacks systematic empirical comparison [5].

As a result, few studies have systematically evaluated reusable bias detection and mitigation pipelines across application domains.

To address this gap, we propose a reusable fairness evaluation and mitigation pipeline and validate its effectiveness on public datasets from the healthcare and education domains. Based on student performance and depression risk prediction tasks, we investigate the following research questions: RQ1: Do machine learning models exhibit systematic predictive disparities across demographic groups in student performance and depression risk prediction tasks? RQ2: Which fairness metrics can consistently detect and quantify group bias across different datasets and domains? RQ3: Which model-agnostic mitigation techniques effectively reduce group bias while preserving predictive performance?

II. Related Work

A. Fairness in Machine Learning

Several survey studies have summarized fundamental concepts, evaluation metrics, and methodological frameworks in machine learning fairness research [1], providing a macro-level overview of fairness definitions, sources of bias, and mitigation strategies. In decision-making contexts, fairness generally refers to the principle that decisions should not exhibit bias or undue favoritism based on the inherent or acquired characteristics of individuals or groups [1]. When algorithmic decisions benefit or harm particular groups—especially those defined by protected attributes such as race, gender, age, ethnicity, or religion—the system is considered unfair [1,6]. Real-world cases demonstrate that algorithmic decisions can exhibit racial or gender bias across different domains, from criminal risk prediction to hiring and service allocation [18,19,20].

Existing research typically categorizes sources of bias in machine learning into two types: data bias and algorithmic bias [1]. Data bias arises from imbalances or historical biases in training data, while algorithmic bias may further amplify these imbalances during model training and optimization. To quantify decision disparities across different groups, researchers have proposed various fairness evaluation metrics. Based on these metrics, bias mitigation methods have been developed for different stages of the machine learning lifecycle, including pre-processing, in-processing, and post-processing strategies [1]. Pre-processing techniques mitigate bias by adjusting data before model training; in-processing techniques try to modify and change the ML algorithm; post-processing techniques adjust model predictions after training is complete. Based on these methods, various tools and frameworks have been developed to support systematic fairness analysis and experimental evaluation. Among them, AI Fairness 360 (AIF360) integrates multiple fairness metrics and mitigation algorithms, providing a unified experimental platform for fairness research across models and datasets [17].

B. Fairness in Healthcare Applications

In healthcare applications, fairness concerns were initially identified through performance disparities across population groups. Prior studies show that even when overall predictive performance is high, models may still exhibit systematic imbalances in decision-making across gender, racial, or ethnic groups. For example, in disease progression prediction tasks based on longitudinal data from the Alzheimer's Disease Neuroimaging Initiative (ADNI), research has found that when data are relatively balanced along the gender dimension, models tend to demonstrate better gender fairness, while substantial disparities persist across racial and ethnic groups [6]. This indicates that structural imbalances in medical data may lead to persistent bias against specific populations, and much of the existing work remains focused on bias identification rather than mitigation.

As fairness issues gradually emerged, subsequent research began to explore mitigation strategies in healthcare prediction tasks. In mental health disorder analysis, some studies have conducted systematic comparisons of pre-processing, in-processing, and post-processing mitigation methods using multimodal datasets that include motor activity and AV information, while employing a broader set of fairness metrics for evaluation [3]. These studies suggest that model bias may arise from both data distribution imbalances and algorithmic design choices. Fairness mitigation methods at different stages exhibit significant variations in their impact on fairness enhancement and predictive performance. Furthermore, some approaches may improve fairness for minority groups while adversely affecting the interests of majority groups.

Furthermore, researchers proposed a unified evaluation pipeline, FAIR-CARE, for systematically comparing the effectiveness of different fairness mitigation methods in medical prediction tasks [16]. Experimental results indicate that various mitigation approaches involve different trade-offs between fairness and predictive performance, which highlights the complexity of fairness interventions in medical settings.

To improve the applicability of research findings in real-world healthcare environments, some studies have further extended fairness analysis to multi-country and multi-population datasets, comparing the performance of mitigation methods across different medical contexts and proposing new approaches such as PSTA [4]. These findings indicate that the same models and mitigation methods may exhibit different fairness behaviors across populations, and in certain cases can effectively reduce bias without sacrificing predictive accuracy. These results demonstrate that medical fairness research should extend beyond single prediction tasks and evaluate the stability of fairness methods across various populations and datasets.

Beyond algorithmic improvements, researchers have also begun examining fairness issues from the perspective of training data. One study introduced the "debugging" principles from software engineering into ML fairness research. By identifying bias features in training data that are highly correlated with sensitive attributes, it corrects the data during the preprocessing stage to enhance fairness. The study also proposed using the "favorable decision rate" metric to evaluate the practical feasibility of fairness methods in real-world applications, highlighting that fairness improvements in medical settings should not overlook potential social resource costs [21].

In research that is closer to real-world applications, some researchers attempt to construct reusable ML pipelines that integrate fairness analysis and mitigation strategies into complete healthcare prediction workflows, applying them to specific tasks to reduce health disparities among disadvantaged groups [5]. However, such approaches often rely on particular models or mitigation techniques and lack systematic validation on other datasets or application domains. As a result, their transferability across tasks and domains requires further investigation.

Overall, fairness research in healthcare has evolved from bias identification to comparative evaluations of mitigation methods, with increasing attention to multi-dataset, multi-population settings and real-world application costs. Nevertheless, most existing studies focus on task-specific or domain-tailored solutions, and systematic empirical investigations of model-agnostic and cross-domain transferable fairness pipelines remain limited.

C. Fairness in Educational Applications

While fairness research has been extensively explored in healthcare applications, similar concerns have increasingly emerged in educational prediction systems. The increasing adoption of machine learning models in the field of education and data mining has given rise to advanced predictions to calculate student outcomes, for example, their performance, risk of dropping out, and engagement. Recent works highlight that machine learning models may be biased toward certain social and demographic groups, potentially leading to unfair treatment of some students. One of the most popular benchmarks for the analysis of student behavior and performance in online learning environments was introduced by Kuzilek et al. [9] in the form of the Open University Learning Analytics Dataset (OULAD). The dataset includes demographic data, course design, assessment data, and highly detailed virtual learning environment interaction data from more than thirty thousand students, allowing for the analysis of learning behavior and performance on a large scale. The inclusion of both demographic and behavioral attributes makes OULAD particularly suitable for fairness analysis,The inclusion of both demographic and behavioral attributes makes OULAD particularly suitable for fairness analysis, as it enables researchers to examine performance differences across demographic groups while considering both behavioural and contextual factors [9].

Early educational data mining research primarily emphasized predictive performance, with limited consideration of fairness issues [10]. Romero and Ventura's survey [10] similarly highlights the strong emphasis on predictive models such as logistic regression, decision trees, and ensemble methods for forecasting student grades and dropout rates. Few studies examined whether these models treated different demographic groups equitably. The attention to accuracy explains why fairness in learning analytics has emerged as an important research focus in recent years.

Baker and Hawn [11] take a closer look at algorithmic bias in educational systems. They argue that models trained on historical educational data risk perpetuating existing inequalities. Bias doesn't just seep in from the data itself, but from the choices which are made while building and using the model. Baker and Hawn insist that fairness checks need to be part of the evaluation process, not an afterthought. Dwork and colleagues [12] go a step further with their idea of individual fairness. They propose the principle of individual fairness, which requires that similar individuals receive similar outcomes. Then Hardt et al. [13] layer on equality of opportunity: everyone should see the same chance for a true positive, no matter what group they're in.These theoretical frameworks have directly influenced the development of modern fairness metrics and post-processing techniques implemented in contemporary toolkits.

Recently, the focus has shifted to fairness in student performance prediction itself. Le Quy et al. [22] examined group fairness metrics in these models and found that fairness outcomes can vary substantially across demographic groups, depending on which fairness measure you use. Their work makes it obvious: accuracy alone isn't enough;Their results demonstrate the importance of carefully selecting appropriate fairness metrics. Marras et al. [23] further investigated bias mitigation strategies by comparing demographic-based and behavioral-based oversampling approaches. Their findings indicate that data-level interventions can reduce disparities, although such improvements may involve trade-offs with predictive performance.Their findings indicate that data-level interventions can reduce disparities, although such improvements may involve trade-offs with predictive performance.

Recent research has also examined alternative formulations of fairness. Kusner et al. [14] introduced the concept of counterfactual fairness, which evaluates whether a model's prediction would remain unchanged if a sensitive attribute were hypothetically altered while all other variables were held constant. This causal perspective enables the assessment of whether sensitive attributes directly influence model outcomes. In addition, Barocas, Hardt, and Narayanan [15] provide a systematic framework for defining, measuring, and mitigating fairness in algorithmic systems, establishing widely adopted conceptual and methodological foundations for fairness research.

Other research warns that sticking with traditional accuracy-focused modeling makes it easy to overlook big differences in how models treat different groups. Collectively, these studies suggest that models should not be evaluated solely on overall predictive accuracy, but also on their fairness across demographic groups.

Despite these advancements, fairness in educational prediction remains an open research challenge. Many existing studies focus on specific tasks or isolated mitigation strategies, and systematic cross-domain evaluations remain limited. This motivates the need for structured fairness evaluation and mitigation pipelines that can be consistently applied across different datasets and application contexts.

D. Comparative Summary

Given the research gaps mentioned above, our main contributions can be summarized as follows. First, we propose

TABLE I
COMPARISON OF FAIRNESS-RELATED STUDIES IN HEALTHCARE AND OUR PROPOSED APPROACH

Study	**Medical Task**	**Multi-dataset**	**Mitigation Methods**	**Fairness–Performance Trade-off**	**Model-agnostic**	**Cross-domain Validation**	**Pipeline Integration**
[6]	AD progression prediction	×	×	×	✓	×	×
[3]	Depression and schizophrenia prediction	✓	✓	×	✓	×	×
[16]	Multiple medical prediction tasks	✓	✓	✓	✓	×	×
[4]	Depression prediction	✓	✓	✓	✓	×	×
[21]	Non-specific medical tasks	✓	✓	✓	✓	✓	×
[5]	Hospital readmission prediction for diabetic patients	×	✓	×	×	×	✓
Ours	Depression and student performance prediction	✓	✓	✓	✓	✓	✓

TABLE II
COMPARISON OF FAIRNESS-RELATED STUDIES IN EDUCATIONAL PREDICTION MODELS AND OUR PROPOSED APPROACH

Study	**Dataset**	**Fairness Analysis**	**Mitigation Methods**	**Fairness–Performance Trade-off**	**Model-agnostic**	**Cross-domain Validation**	**Pipeline Integration**
Kuzilek et al. (2017)	OULAD	×	×	×	×	×	×
Baker & Hawn (2021)	Education systems	✓	×	✓	✓	×	×
Le Quy et al. (2022)	Student performance	✓	×	✓	✓	×	×
Marras et al. (2023)	Student success	✓	✓	✓	×	×	×
Kusner et al. (2017)	General ML	✓	×	✓	✓	×	×
Barocas et al. (2020)	General ML	✓	✓	✓	✓	✓	×
Hardt et al. (2016)	General ML	✓	✓	✓	✓	✓	×
Dwork et al. (2012)	General ML	✓	Conceptual	×	✓	✓	×
Romero & Ventura (2010)	Educational data	×	×	×	×	×	×
Kesgin et al. (2025)	Student performance	✓	✓	✓	×	×	×
Ours	OULAD and NHANES	✓	✓	✓	✓	✓	✓

a general and model-agnostic machine learning pipeline for classification tasks, enabling systematic comparison of different models and fairness methods within a unified framework. Second, we integrate fairness evaluation metrics with multiple fairness mitigation strategies into the pipeline, allowing for a structured analysis of the trade-offs between fairness improvement and predictive performance. Finally, we apply the proposed pipeline to cross-domain datasets to assess the stability and consistency of fairness metrics and mitigation strategies across application scenarios, and to identify approaches that demonstrate greater reliability in cross-domain settings. Tables 1 and 2 summarize the main differences between prior work and our proposed approach in the healthcare and education domains, respectively.

ACKNOWLEDGMENT

In preparing this literature review, the author consulted artificial intelligence tools (e.g., ChatGPT) only to clarify certain concepts and refine language where necessary. All literature retrieval, material organization, viewpoint development, and final writing were completed independently by the author.

REFERENCES

[1] N. Mehrabi, F. Morstatter, N. Saxena, K. Lerman, and A. Galstyan, "A Survey on Bias and Fairness in Machine Learning," *ACM Comput. Surv.*, vol. 54, no. 6, pp. 115:1–115:35, Jul. 2021. [Online]. Available: https://dl.acm.org/doi/10.1145/3457607

[2] "The Act Texts | EU Artificial Intelligence Act." [Online]. Available: https://artificialintelligenceact.eu/the-act/

[3] J. Cheong, S. Kuzucu, S. Kalkan, and H. Gunes, "Towards Gender Fairness for Mental Health Prediction." International Joint Conferences on Artificial Intelligence Organization, Aug. 2023. [Online]. Available: https://www.repository.cam.ac.uk/handle/1810/349873

[4] V. N. Dang, A. Cascarano, R. H. Mulder, C. Cecil, M. A. Zuluaga, J. Hernández-González, and K. Lekadir, "Fairness and bias correction in machine learning for depression prediction across four study populations," *Scientific Reports*, vol. 14, no. 1, p. 7848, Apr. 2024. [Online]. Available: https://www.nature.com/articles/s41598-024-58427-7

[5] S. Raza, "A machine learning model for predicting, diagnosing, and mitigating health disparities in hospital readmission," *Healthcare Analytics*, vol. 2, p. 100100, Nov. 2022. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2772442522000430

[6] C. Yuan, K. A. Linn, and R. A. Hubbard, "Algorithmic Fairness of Machine Learning Models for Alzheimer Disease Progression," *JAMA Network Open*, vol. 6, no. 11, p. e2342203, Nov. 2023. [Online]. Available: https://doi.org/10.1001/jamanetworkopen.2023.42203

[7] Q. Feng, M. Du, N. Zou, and X. Hu, "Fair Machine Learning in Healthcare: A Survey," *IEEE Transactions on Artificial Intelligence*, vol. 6, no. 3, pp. 493–507, Mar. 2025. [Online]. Available: https://ieeexplore.ieee.org/document/10700762/metrics

[8] R. R. Fletcher, A. Nakeshimana, and O. Olubeko, "Addressing Fairness, Bias, and Appropriate Use of Artificial Intelligence and Machine Learning in Global Health," *Frontiers in Artificial Intelligence*, vol. 3, Apr. 2021. [Online]. Available: https://www.frontiersin.org/journals/artificial-intelligence/articles/10.3389/frai.2020.561802/full

[9] J. Kuzilek, M. Hlosta, and Z. Zdrahal, "Open university learning analytics dataset," *Scientific data*, vol. 4, no. 1, p. 170171, 2017.

[10] C. Romero and S. Ventura, "Educational data mining and learning analytics: An updated survey," *Wiley interdisciplinary reviews: Data mining and knowledge discovery*, vol. 10, no. 3, p. e1355, 2020.

[11] R. S. Baker and A. Hawn, "Algorithmic bias in education," *International journal of artificial intelligence in education*, vol. 32, no. 4, pp. 1052–1092, 2022.

[12] C. Dwork, M. Hardt, T. Pitassi, O. Reingold, and R. Zemel, "Fairness through awareness," in *Proceedings of the 3rd innovations in theoretical computer science conference*, 2012, pp. 214–226.

[13] M. Hardt, E. Price, and N. Srebro, "Equality of opportunity in supervised learning," *Advances in neural information processing systems*, vol. 29, 2016.

[14] M. J. Kusner, J. Loftus, C. Russell, and R. Silva, "Counterfactual

fairness," *Advances in neural information processing systems*, vol. 30, 2017.

[15] S. Barocas, M. Hardt, and A. Narayanan, "Fairness and machine learning," 2019.

[16] C. Criscuolo, M. Salnitri, and D. Martinenghi, "FAIR-CARE: A comparative evaluation of unfairness mitigation approaches," *Information and Software Technology*, vol. 189, p. 107898, Jan. 2026. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S095058492500237X

[17] R. K. E. Bellamy, K. Dey, M. Hind, S. C. Hoffman, S. Houde, K. Kannan, P. Lohia, J. Martino, S. Mehta, A. Mojsilović, S. Nagar, K. N. Ramamurthy, J. Richards, D. Saha, P. Sattigeri, M. Singh, K. R. Varshney, and Y. Zhang, "AI Fairness 360: An extensible toolkit for detecting and mitigating algorithmic bias," *IBM Journal of Research and Development*, vol. 63, no. 4/5, pp. 4:1–4:15, Jul. 2019. [Online]. Available: https://ieeexplore.ieee.org/document/8843908

[18] J. Angwin, J. Larson, S. Mattu, and L. Kirchner, "Machine Bias," May 2016. [Online]. Available: https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing

[19] J. Dastin, "Amazon scraps secret AI recruiting tool that showed bias against women," Oct. 2018, section: In The Workplace. [Online]. Available: https://www.independent.ie/business/in-the-workplace/amazon-scraps-secret-ai-recruiting-tool-that-showed-bias-against-women/37403938.html

[20] C. O'Donovan and A. Gregg, "Amazon deliveries are slower in low-income D.C. Zip codes, lawsuit says," *The Washington Post*, Dec. 2024. [Online]. Available: https://www.washingtonpost.com/technology/2024/12/04/amazon-deliveries-slow-lawsuit-dc/

[21] Y. Li, L. Meng, L. Chen, L. Yu, D. Wu, Y. Zhou, and B. Xu, "Training data debugging for the fairness of machine learning software," in *Proceedings of the 44th International Conference on Software Engineering*, ser. ICSE '22. New York, NY, USA: Association for Computing Machinery, Jul. 2022, pp. 2215–2227. [Online]. Available: https://dl.acm.org/doi/10.1145/3510003.3510091

[22] T. Le Quy, T. H. Nguyen, G. Friege, and E. Ntoutsi, "Evaluation of group fairness measures in student performance prediction problems," in *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*. Springer, 2022, pp. 119–136.